

Lab: Health Planning -- Your Body's Prediction Machine

Objective

Learning Goal: Investigate how your body plans ahead through immune memory, sleep cycles, circadian rhythms, and how you can use prediction-based thinking to plan for better health.

This is an **investigation lab** designed for middle school students (Grades 6-8).

Materials Needed

- Paper and colored pencils
 - Health tracker template (provided below)
 - Lab journal or notebook
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Part 1: Your Body's Built-In Plans (10 min)

Your body does not just react -- it plans ahead. Explore these examples of biological planning.

Body Planning System	What does it plan for?	How does it use prediction?	What happens when the plan fails?
Immune memory (vaccines)	Future infections	Remembers past invaders to prepare	
Circadian rhythm (sleep cycle)	Day/night transitions	Predicts when to feel sleepy/alert	
Blood sugar regulation	Energy needs	Predicts when food is coming	
Seasonal changes (mood, energy)	Changes in daylight	Predicts environmental shifts	

Write your response here

Part 2: Circadian Rhythm Tracking (15 min)

Your body has a built-in clock that predicts when you should be awake, asleep, hungry, and alert. Map your personal circadian rhythm.

Time of Day	Energy Level (1-10)	Hunger Level (1-10)	Alertness (1-10)	Best activity for this time
7:00 AM				
10:00 AM				
12:00 PM				
2:00 PM				
4:00 PM				
7:00 PM				
10:00 PM				

Write your response here

Part 3: Design a Health Plan (10 min)

Using what you know about how your body predicts and plans, design a one-week health plan for yourself.

Health Element	Current Habit	Better Prediction-Based Plan	Why this helps your body's predictions
Sleep schedule			
Meal timing			
Exercise			
Screen time before bed			
Stress management			

Write your response here

Part 4: Reflection Table

Question	Your Answer
How does your body plan ahead without you thinking about it?	
What is a circadian rhythm and how does it use prediction?	
How does immune memory work as a biological plan?	
Why does a regular schedule help your body?	
How does Active Inference explain health planning?	

Write your response here